

On the Status of Local Entropy-Current Extensions of the Gravitational Entropy Escrow Framework: A Clarification Note

Grant L. Whitmer III

Windstorm Institute, Fort Ann, New York

Draft v0.3.1 — May 2026

Abstract

In dialogue exploring possible covariant extensions of the Gravitational Entropy Escrow framework (Whitmer 2026), several large-language-model systems independently proposed an equation of the form $\nabla_\mu J_S^\mu = (2\pi k_B / \hbar) \cdot \varepsilon_{\text{bind}}$, designated C8, in which a local entropy 4-current is sourced by the local Tolman-Komar binding-energy density. We document the resolution of this proposal across multiple rounds of analysis. With dimensions accounted for correctly, with the standard energy density of the cosmological constant $\varepsilon_\Lambda = \Lambda c^4 / (8\pi G)$, and with the integration time chosen as the light-crossing time of the relevant horizon, C8 reproduces both the Bekenstein-Hawking entropy of a Schwarzschild black hole and the Gibbons-Hawking entropy of the de Sitter cosmological horizon exactly. However, this recovery is a consistency check rather than a derivation: C8 with $t = R/c$ is algebraically identical to the saturated Bekenstein bound (Bekenstein 1981), and any horizon that saturates the Bekenstein bound by construction will be reproduced by C8 trivially. The choice of integration time $t = R/c$ is post-hoc; other natural time scales associated with the horizon (Hawking evaporation, inverse surface gravity, free-fall) give different answers ranging across many orders of magnitude. C8 therefore reduces to a rate-equation reformulation of an established 1981 result rather than producing genuinely new physics. We document this resolution — and the chain of dimensional and convention errors that complicated reaching it — to spare future researchers extending the escrow framework from the same dead end. The published paper's interpretive content is unaffected; its central open problem (the order-unity coefficient $\alpha \approx 1.34$ between the bare Λ -form a_0 and the empirical Milgrom value) is not addressed by C8 in either direction.

1. Motivation

The Gravitational Entropy Escrow framework (Whitmer 2026, henceforth W26) proposes that gravitational binding energy is structurally identical to entropy held in escrow against the local Unruh temperature: $S_{\text{esc}} = |U_{\text{grav}}| / T$, with the force law $F = T \cdot \nabla S_{\text{esc}}$ recovering Newton's law (Verlinde 2011). The framework recovers Bekenstein-Hawking entropy at Schwarzschild horizons, the order of magnitude of the empirical Milgrom acceleration scale a_0 from a thermodynamic crossover at the de Sitter horizon, and several other results discussed in W26. Section 7.6 of W26 identifies a complete derivation of modified field equations from a variational principle on total entropy as the natural next step. In dialogue with multiple large-language-model systems exploring such an extension, a covariant equation of the form

$$\nabla_\mu J_S^\mu = (2\pi k_B / \hbar) \cdot \varepsilon_{\text{bind}} \quad [C8]$$

was independently proposed, with J_S^μ an entropy 4-current and $\varepsilon_{\text{bind}}$ the Tolman-Komar binding-energy density. The dimensions balance: both sides have units of entropy production density per unit time, $k_B / (m^3 \text{ s})$.

This note documents the resolution of C8's status. The result is small and specific: C8 with the light-crossing-time integration prescription is algebraically identical to the saturated Bekenstein bound (Bekenstein 1981). It therefore reproduces both the Bekenstein-Hawking and Gibbons-Hawking entropies exactly when the standard energy density of the cosmological constant is used. The recovery is a consistency check rather than a derivation, because horizons saturate the Bekenstein bound by construction. The choice of integration time is post-hoc rather than derived. C8 is therefore not a viable independent extension of W26; it is a rate-equation reformulation of a 1981 result.

This note also documents, briefly, a sequence of errors made during the analysis. We include them not because the errors are interesting in themselves but because they are easy to repeat, and the methodological lesson is sharper than the technical one. The companion methodology paper (in preparation) will treat this case study at greater length.

2. C8 as a Rate Equation

C8 is a continuity-like equation in which the divergence of an entropy 4-current is sourced by an energy density. Both sides have dimensions of entropy production density per unit time. To compare C8 to a total entropy (such as Bekenstein-Hawking or Gibbons-Hawking) one must first integrate the rate over both volume and time.

Integrating C8 over a region of volume V containing total binding energy E gives the total entropy production rate for the region:

$$dS/dt = (2\pi k_B / \hbar) \cdot E$$

Multiplying by a characteristic time t_h yields a total entropy:

$$S_{C8} = (2\pi k_B / \hbar) \cdot E \cdot t_h$$

The choice of t_h is the load-bearing question. It is not specified by C8 itself.

3. C8 with $t = R/c$ is the Saturated Bekenstein Bound

Choosing $t_h = R/c$, where R is the radius of the region:

$$S_{C8} = (2\pi k_B / \hbar) \cdot E \cdot (R/c) = 2\pi k_B E R / (\hbar c)$$

This is the Bekenstein bound (Bekenstein 1981), saturated. The bound applies as an inequality to any localized system: the maximum entropy that a region of energy E and radius R can contain in a quantum field theory respecting causality and the second law is $2\pi k_B E R / (\hbar c)$. Black hole event horizons saturate the bound, with the saturation value being the Bekenstein-Hawking entropy. The de Sitter cosmological horizon also saturates the bound when the appropriate energy is the total vacuum energy enclosed within the horizon.

Therefore C8 with light-crossing-time integration equals the saturated Bekenstein bound by algebra, and reproduces horizon entropies for any horizon that saturates the bound. This is a known relation written in a different form, not new physics.

4. Recovery of Horizon Entropies

4.1 Schwarzschild black hole

For a Schwarzschild black hole of mass M , taking the rest energy $U = Mc^2$ and the Schwarzschild radius $r_s = 2GM/c^2$ as the relevant radius:

$$S_{C8}(BH) = (2\pi k_B / \hbar) \cdot (Mc^2) \cdot (r_s/c) = 4\pi G M^2 k_B / (\hbar c) = S_{BH}$$

The ratio $S_{C8(BH)} / S_{BH} = 1$ exactly, independent of M . This is verified numerically and symbolically in the accompanying script.

4.2 de Sitter cosmological horizon

For the de Sitter horizon, the radius is $R_{dS} = \sqrt{3/\Lambda}$, a length (units m). Using the standard energy density of the cosmological constant:

$$\varepsilon_{\Lambda} = \Lambda c^4 / (8\pi G) \quad [\text{units: J/m}^3]$$

This is the energy density ($\approx 5.3 \times 10^{-10} \text{ J/m}^3$ for our universe, in agreement with Planck 2018). The closely related quantity $\Lambda c^2/(8\pi G)$ is the equivalent mass density ($\approx 6.0 \times 10^{-27} \text{ kg/m}^3$) and must not be substituted for the energy density in this calculation, as we discuss in Section 7. The total vacuum energy enclosed within R_{dS} is $E_{\text{total}} = \varepsilon_{\Lambda} \cdot V_{dS}$ where $V_{dS} = (4\pi/3)R_{dS}^3$. Then:

$$S_{C8(dS)} = (2\pi k_B / \hbar) \cdot (\varepsilon_{\Lambda} V_{dS}) \cdot (R_{dS}/c) = 3\pi c^3 k_B / (G \hbar \Lambda) = S_{GH}$$

The ratio $S_{C8(dS)} / S_{GH} = 1$ exactly, independent of Λ . This is verified in the accompanying script.

5. The Light-Crossing Time is Post-Hoc

The choice of $t_h = R/c$ is not derived from C8. It is one of several time scales naturally associated with a horizon. For a Schwarzschild black hole of solar mass, alternative natural choices include:

- Light-crossing time $r_s/c \approx 10^{-5} \text{ s}$, giving ratio $S_{C8}/S_{BH} = 1$.
- Inverse surface gravity $c/\kappa \approx 2 \times 10^{-5} \text{ s}$, giving ratio = 2.
- $\pi r_s/c$ (related to the inverse Hawking frequency) $\approx 3 \times 10^{-5} \text{ s}$, giving ratio = π .
- Hawking evaporation time $\tau_{\text{evap}} \approx 10^{64} \text{ yr}$, giving ratio $\approx 10^{80}$.

Only the light-crossing time produces ratio = 1, and it does so because R/c is precisely the factor needed for C8 with prefactor $(2\pi k_B / \hbar)$ to algebraically equal the Bekenstein bound $2\pi k_B E R/(\hbar c)$. The other natural time scales — each defensible on its own physical grounds — give different answers ranging across many orders of magnitude.

This is the textbook signature of a post-hoc choice. A genuine derivation would either select t_h from a deeper principle (e.g., the Raychaudhuri equation, null geodesic focusing, or detailed balance on the horizon) or would survive without privileging any particular choice. C8 with R/c does neither. The agreement with horizon entropies that R/c produces is therefore a consequence of the algebraic identity between $(2\pi k_B/\hbar) \cdot E \cdot (R/c)$ and the Bekenstein bound, not independent evidence for C8.

6. Status and Implications

C8 with the light-crossing-time prescription is algebraically equivalent to the saturated Bekenstein bound. It therefore reproduces Bekenstein-Hawking and Gibbons-Hawking entropies for any horizon that saturates the bound, but this is a consistency check on dimensional structure rather than independent confirmation of C8 as a covariant extension of the escrow framework. C8 contains no information beyond what is already in Bekenstein 1981 plus the choice of integration time.

The published escrow framework (W26) is unaffected by this analysis. Its interpretive content does not depend on C8. Specifically:

- The static escrow postulate $S_{\text{esc}} = |U_{\text{grav}}|/T$ at any configuration is unaffected.
- The recovery of Newton's law via Verlinde's identity is unaffected.

- The recovery of Bekenstein-Hawking entropy at horizons (W26 Section 5) is unaffected; that recovery is independent of C8 and follows from substituting the Schwarzschild surface gravity into the static escrow expression.
- The identification of the deep-MOND limit with a thermodynamic crossover at the de Sitter horizon (W26 Section 6) is unaffected.
- The empirical tests reported in W26 Sections 6.1, 8.4, 8.5, 8.6 (the Genzel five-case test, the SPARC reanalysis, the baryonic Tully-Fisher relation, and cross-redshift consistency) are all unaffected because they depend on the deep-MOND asymptote $a_0 = c^2\sqrt{(\Lambda/3)}/(2\pi)$, not on C8.

What is ruled out by this analysis is the natural-looking local-current generalization of the static escrow postulate as a candidate independent extension of W26. Future researchers seeking a covariant extension of W26 should be aware that this direction does not yield new content beyond the static postulate plus Bekenstein 1981.

Three directions for future extension of W26 are not ruled out by this analysis:

- Quantum corrections to horizon temperatures. The leading-order Gibbons-Hawking and Hawking temperatures may receive corrections from entanglement-entropy structure across horizons, which depend on the cosmological matter content (Ω_m) or other parameters. Such corrections might derive the order-unity coefficient $\alpha \approx 1.34$ named as the open problem in W26 Section 8.4.3. Calculations of this type have been done in lower-dimensional analogs (Calabrese-Cardy 2004) but not, to our knowledge, for the four-dimensional de Sitter horizon in a universe with realistic matter content.
- Rindler–de Sitter horizon matching. The framework's commitment to constant a_0 (W26 Section 6) requires the de Sitter horizon temperature to set the floor for entropy exchange of accelerated observers within the universe. The matching condition between local Rindler horizons and the global de Sitter horizon is a structural problem the existing paper does not resolve; a correct treatment would not be a local continuity equation.
- Lattice quantum field theory tests of the static escrow postulate. The identification $S_{\text{esc}} = |U_{\text{grav}}|/T$ can be tested directly by computing the entanglement entropy across a boundary separating two localized mass distributions in a quantum field theory on a lattice and checking whether the result scales as $|U_{\text{grav}}|/T_{\text{Unruh}}$. This is computationally tractable in 1+1D using Peschel's formula for Gaussian states (Peschel 2003) and remains a sharp falsification test of the static postulate.

The framework's central open derivation problem ($\alpha \approx 1.34$) lives in the deep-MOND interpolation, not in horizon entropy magnitudes. This note's analysis of C8 does not address it in either direction.

7. Methodological Notes

This paper is the third draft of an analysis of C8. Two earlier drafts reported wrong results for opposite reasons. We document the chain of errors so that future researchers — human or AI — can recognize the failure modes.

The first draft (v0.1) reported that C8 fails by 30 orders of magnitude when applied to galactic systems and by 4.7 orders of magnitude for solar-mass black holes. This was a dimensional category error: the C8 source term (an entropy production rate density) was compared directly to total entropies. The category error was not detectable from numerical magnitudes alone because the apparent failure (30 orders of magnitude) was large enough to seem qualitatively diagnostic regardless of dimensional details. The error would have been caught immediately by a sympy-based dimensional check, which we did not perform on the first pass.

The second draft (v0.2) corrected the rate-vs-total error and re-evaluated C8 with $t = R/c$ integration. It reported that C8 reproduces Bekenstein-Hawking exactly but underpredicts Gibbons-Hawking by $1/c^2 \approx 10^{-17}$. This was a different kind of error: the calculation used the

mass density $\Lambda c^2/(8\pi G)$ where the energy density $\Lambda c^4/(8\pi G)$ was required. The two differ by exactly c^2 , which propagates as a factor of c^2 in the de Sitter total energy and therefore in S_{C8} , producing a spurious "underprediction by c^2 " result. Symbolic dimensional analysis cannot catch this kind of error because both forms are dimensionally consistent within their respective conventions; only physical identification ("this should be an energy density, not a mass density, because we are computing a total energy not a total mass") catches it.

The third draft (this document) compares each computed quantity against a published reality-check value before treating it as established. The dark energy density $\epsilon_\Lambda = 5.3 \times 10^{-10} \text{ J/m}^3$ is published in Planck 2018; the de Sitter horizon radius $R_{dS} \approx 16 \text{ Gly}$ is published in standard cosmology references; the Gibbons-Hawking entropy $S_{GH} \approx 10^{122} k_B$ for our universe is published in horizon-entropy reviews. Computing each from first principles and comparing against the published value catches convention errors that dimensional analysis does not. After Perplexity's audit identified the mass-vs-energy density error, a fourth round of review was sought from xAI Grok in the form of an "audit of the audit." Grok's response confidently rejected Perplexity's correction and asserted that $\Lambda c^2/(8\pi G)$ is the correct energy density of the cosmological constant in SI units, attributing the disagreement to Perplexity having confused mass and energy density conventions. This is the opposite of what dimensional analysis and the published Planck 2018 values show: $\Lambda c^2/(8\pi G)$ yields $5.93 \times 10^{-27} \text{ kg/m}^3$ (matching the published mass density) and $\Lambda c^4/(8\pi G)$ yields $5.33 \times 10^{-10} \text{ J/m}^3$ (matching the published energy density). Grok's audit was confidently wrong in the same direction that the v0.2 draft had been wrong. Resolving the disagreement required independent first-principles calculation against three sources: dimensional analysis from the SI unit definitions, comparison against Planck 2018 dark-energy values, and cross-check via the Friedmann equation $H^2 = \Lambda c^2/3$ (SI form). All three confirmed Perplexity's correction.

Three methodological lessons follow:

- Symbolic dimensional analysis is necessary but not sufficient. It catches dimensional inconsistency within an assumed convention; it does not catch errors in choosing the right convention. The factor of c^2 between mass density and energy density of the cosmological constant is a case in point: both are dimensionally consistent within their own conventions, but only one is correct for a given physical question.
- Reality checks against published values are necessary. Each derived quantity should be cross-validated numerically against a published value from an independent source before being treated as established. This is the check that catches convention errors and unit conversions that pure mathematics cannot.
- Multi-LLM adversarial review is load-bearing but cannot resolve disagreements among LLMs. When two LLM systems give opposite verdicts on a technical question, the resolution must come from first-principles calculation against external published values, not from a third LLM. In the present case, three of the four systems consulted (Claude, Grok, and an earlier Gemini round) were confidently wrong about the c^2 convention at various points; only Perplexity caught the actual error correctly, and Grok's subsequent "audit of the audit" reversed the correct result back to the wrong answer with high confidence. Recursion of LLM review without external grounding is unstable.

Each of these lessons emerged from this specific analysis. The pattern across rounds — confidently wrong, then confidently right, then confidently wrong again — is characteristic of working with dimensional and convention questions in AI dialogue without consistent external anchoring. The methodology paper in preparation will treat this case study at greater length.

8. Conclusion

C8 is the saturated Bekenstein bound (Bekenstein 1981) expressed as a local rate equation. With the light-crossing-time integration prescription and the standard energy density of the

cosmological constant, it reproduces Bekenstein-Hawking and Gibbons-Hawking entropies exactly. This recovery is a consistency check rather than a derivation, because both horizons saturate the Bekenstein bound by construction. The choice of integration time is post-hoc; other natural choices give different answers across many orders of magnitude. C8 therefore does not constitute an independent covariant extension of the escrow framework, even though it is internally consistent.

The published escrow framework (Whitmer 2026) is unaffected by this result. Its interpretive content stands; its central open problem ($\alpha \approx 1.34$ in the deep-MOND interpolation) is not addressed. Future researchers extending the framework should be aware that the natural-looking local-current generalization reduces to a known 1981 result and should pursue extensions through quantum corrections to horizon temperatures, Rindler–de Sitter matching, or lattice QFT tests of the static postulate, none of which are ruled out by this analysis.

Acknowledgments

The candidate equation C8 was first proposed in dialogue with Anthropic Claude Opus 4.7, with xAI Grok independently proposing an equivalent form. Adversarial review across multiple rounds of analysis was contributed by Anthropic Claude, xAI Grok, Google Gemini, and Perplexity Deep Research. The mass-vs-energy density convention error in the v0.2 draft was identified by Perplexity Deep Research's audit, prompting the corrected v0.3 analysis presented here. A subsequent “audit of the audit” from xAI Grok confidently reversed the correction; resolving this required independent first-principles calculation against Planck 2018 published values, which confirmed Perplexity. The author is solely responsible for the calculations, conclusions, and any remaining errors.

Code and Data Availability

The Python script reproducing all numerical claims (`c8_calculations_v0_3.py`) is deposited alongside this preprint. The script uses both numerical evaluation and symbolic dimensional analysis (`sympy`), and includes explicit reality checks against published values for the dark energy density, the de Sitter horizon radius, and the Gibbons-Hawking entropy. The script runs in under one second and is licensed MIT.

References

- Bekenstein, J. D. (1981). Universal upper bound on the entropy-to-energy ratio for bounded systems. *Physical Review D* 23, 287.
- Calabrese, P. & Cardy, J. (2004). Entanglement entropy and quantum field theory. *Journal of Statistical Mechanics*, P06002.
- Gibbons, G. W. & Hawking, S. W. (1977). Cosmological event horizons, thermodynamics, and particle creation. *Physical Review D* 15, 2738.
- Hawking, S. W. (1975). Particle creation by black holes. *Communications in Mathematical Physics* 43, 199.
- Padmanabhan, T. (2010). Thermodynamical aspects of gravity: New insights. *Reports on Progress in Physics* 73, 046901. arXiv:0911.5004.
- Peschel, I. (2003). Calculation of reduced density matrices from correlation functions. *Journal of Physics A* 36, L205.
- Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics* 641, A6.
- Verlinde, E. P. (2011). On the origin of gravity and the laws of Newton. *JHEP* 04, 029.
- Whitmer, G. L. III (2026). Gravitational Entropy Escrow: An Interpretive Synthesis of Thermodynamic Approaches to Gravity. Zenodo. doi:10.5281/zenodo.20032023.